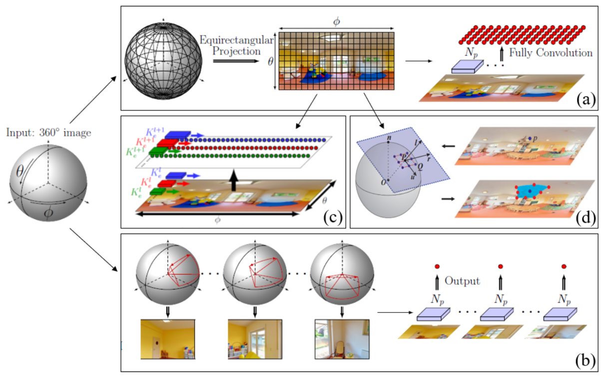

Fig. 11. The principle of CNN for spherical images [66], [67]. (a) direct applying CNN on the ERP spherical images; (b) applying CNN on the tangent images; (c) adjusting the kernel shape of CNN at varying latitudes; (d) adjusting the sampling points of CNN kernels by using tangent plane projection.

tion. According to the work of [35], existing networks can be divided into three groups, i.e., joint feature and metric learning networks [55]–[57], separate detector and descriptor learning networks [58]–[60], [61], and joint detector and descriptor learning networks [62], [63]. For the first group, CNN models learn a similarity function to predict image patch similarities and integrate feature representation and matching within the same network. For the second group, CNN networks only learn feature representation, and feature matching is conducted based on classical strategies, such as the L2-norm Euclidean distance-based metric between feature descriptors. For the third group, trained models learn detectors and descriptors simultaneously, which can cope with images recorded in varying conditions. Similar to hand-crafted methods, existing networks have been used for spherical images by using pre-trained or finetuned models, as shown in Fig. 11(a). For further details, readers can refer to [64], [65].

Due to inevitable distortions in sphere-to-plane projection, CNN networks for perspective images may obtain inaccurate results. To address this issue, reported approaches in literature can be divided into three groups, i.e., tangent projection methods, CNN kernel shape resizing methods, and CNN sampling point adjustment methods.

- For the first one, similar to 2D plane-based methods, equirectangular images are first projected to undistorted tangent images [39] or divided into quasi-uniform discrete images [68], and then existing CNN networks are applied to resulting images, as shown in Fig. 11(b). Although these methods can achieve high accurate predictions, they suffer from high computational costs due to resampling of 3D spherical images to 2D plane images.
- For the second one, CNN networks are designed to work on equirectangular images by adjusting the CNN kernel shape [43], [66], [67]. In the work of [66], a network termed SPHCONV has been proposed, which aims to produce results as the output of applying perspective CNN networks to the corresponding tangent images. SPHCONV was achieved by defining convolution kernels with varying shapes for pixels in different image rows, as illustrated in Fig. 11(c). Similarly, [69] proposed a kernel transformer network (KTN) to learn spherical kernels by taking as input the latitude angle and source kernels for perspective images.
- For the third one, sampling points of CNN kernels are adjusted based on ERP sphere image distortions instead of adjusting convolution kernel shape. For example, [67] and [43] designed distortion-aware networks that sample non-regular grid locations according to the distortions of different pixels, as shown in Fig. 11(d). The core idea is to determine sampling locations based on the sphere projection of a regular grid on the corresponding tangent plane. Due to regular convolution kernels, these frameworks enable the transfer between CNN models for perspective and equirectangular images.

2) Feature matching: The purpose of feature matching is to search correspondences from two sets of feature descriptors. Generally, feature matching is achieved by using nearest-neighbor searching based on the L2-norm Euclidean distance metric [70]. In the literature, extensive research has been reported for feature matching from the aspects of efficiency acceleration and precision improvement [34], [71]. Due to the high dimension of local feature descriptors and the large number of spherical images, exhaustive feature matching consumes extremely high time costs. Thus, it becomes very critical to increase the efficiency of feature matching.

In the literature, there are valuable methods for accelerating feature matching, including restricting the number of features in feature matching, decreasing the number of images, and reducing the number of match pairs. For a detailed review, the readers can refer to this work [34], [72]. Among these reported methods, match pair selection can be the most straightforward and effective way to accelerate feature matching. Three strategies can be exploited for spherical images based on their acquisition environments. First, data acquisition constraints can be used, e.g., the time-sequential constraint for street-view images. In other words, feature matching can be restricted to neighbors according to their acquisition time. Second, for professional MMS systems and consumer-grade sensors, precise or rough GNSS (Global Navigation Satellite System) data is usually recorded simultaneously with spherical images, which provides the best clue to select spatially overlapped match pairs. Third, the CBIR (Content-based Image Retrieval) technique has become a standard module [73], [74]. It merely uses images for visual retrieval without other assumptions and auxiliary data. Fig. 12 shows an example of CBIR.

D. Image orientation

Image orientation aims to estimate camera poses, scene structures, and intrinsic parameters based on established two-view correspondences. Image orientation is also termed aerial triangulation (AT) in photogrammetric 3D reconstruction, which requires good initial values of unknown parameters and well-designed configurations in data acquisition. Image orientation of spherical images can be achieved through the well-known Structure from Motion (SfM) technique for multi-view geometry in computer vision [75] or the Simultaneous

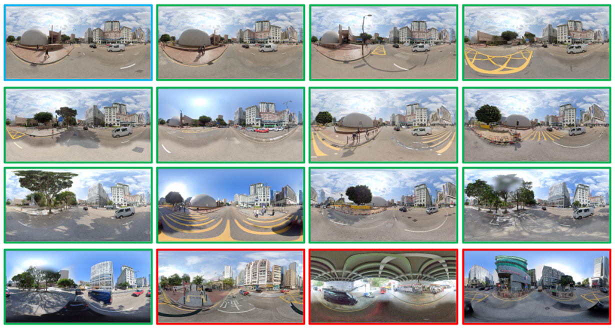

Fig. 12. An example of CBIR for spherical images. The image with blue borders is the query image, and the other images are the retrieval results. Images with green and red borders represent true and false retrieval results, respectively.

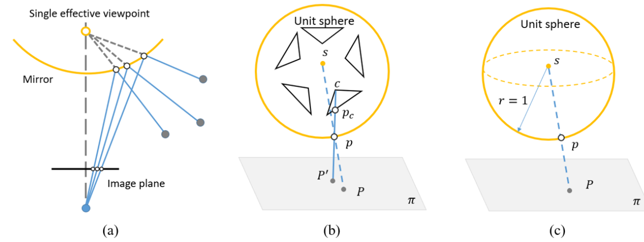

Fig. 13. Camera imaging models [80], [79]. (a) central catadioptric cameras; (b) multi-camera model; and (c) unit sphere camera model.

Localization and Mapping (SLAM) technique for instant localization in robot vision [76]. Thus, this subsection presents camera calibration, SfM-based offline, and SLAM-based online methods for spherical image orientation.

1) Camera calibration:

(1) camera imaging model

The camera imaging model is the basis for camera calibration, which establishes the geometric transformation between 3D points in the object space and 2D points in the image plane. Based on the design of spherical cameras, there are three major camera imaging models, i.e., the unified projection model, general camera model, and multi-camera model.

- **Unified projection model** [77] has been mainly designed for central catadioptric cameras, in which environment light rays intersect in a single point, i.e., the projection center of the mirror, as shown in Fig. 13(a). This camera imaging model follows a strict theoretical projection function that models real-world imaging errors. The unified projection model has recently been verified as effective for wide-angle and fisheye cameras [78].
- **General camera model** (Taylor model) [79] has been used for modeling the imaging procedure of central catadioptric and dioptric cameras. Instead of using the strict theoretical model in the unified projection model, the general camera model utilizes a Taylor polynomial function to fit the projection.
- **Multi-camera model** has been designed to establish the projection of the widely used polydioptric cameras that record spherical images by using a camera rig, e.g., the Ladybug 5+ camera. The multi-camera model can be implemented using several individual camera models or a unit sphere camera model [80]. The former is more rigorous in formulating the imaging system, as shown in Fig. 13(b); on the contrary, the latter has a more straightforward formula widely used in close-range photogrammetry, as shown in Fig. 13(c).

(2) camera calibration

The purpose of camera calibration is to calculate the intrinsic parameters of cameras, e.g., the focal length, principle point, and lens distortion coefficients. After selecting a proper camera imaging model, camera calibration can be achieved by using a combined bundle adjustment of both interior and exterior orientation parameters. In practice, there are three ways to estimate the parameters: self-calibration based on the epipolar geometry constraint, space resection using non-planar 3D control points, and laboratory calibration using checkerboards [81]. In the work of [82], a Ricoh Theta dual-camera system has been calibrated based on the expanded unit sphere camera model, in which two extra parameters that model the displacement of latitude and longitude coordinates are incorporated into the camera imaging model. For calibrating a professional Ladybug multi-camera system, [83] adopted the colinear equation-based rigorous model, which combines five radial distortion parameters and two decentering distortion parameters to model the imaging errors. For comparing different calibration methods for spherical cameras, readers can refer to the work of [84]. Besides, Table IV presents a list of open-source software packages for spherical camera calibration.

2) SfM-based offline methods:

(1) Principle of incremental SfM

Existing SfM can be divided into three major groups, i.e., incremental SfM, global SfM, and hybrid SfM, according to the used strategy for estimating and optimizing unknown parameters [87]. Compared with other techniques, incremental SfM has the advantages of resisting high outlier ratios and achieving accurate orientation models. Incremental SfM has been widely used in photogrammetric 3D reconstruction [49].

The workflow of incremental SfM is shown in Fig. 14, which consists of two components, i.e., feature matching and incremental image registration. Feature matching can be solved by using the methods presented in Section III-A. For image registration, consistent correspondences are first tied to create tie-points [88]. In incremental image registration, two seed images are first selected among all matched image pairs, which have a large enough intersection angle and a sufficient number of well-distributed matched features. A base model is then constructed by recovering their relative poses and triangulating 3D scene points, which would be used to register the next-best image and triangulate more 3D scene points. Meanwhile, local or global bundle adjustment (BA) optimization is executed to decrease accumulated errors and remove false matches. After iterative image registration and point triangulation, all images are registered into the same 3D model [89].

In the above-mentioned iterative local and global BA, the optimization problem is usually formulated as a joint minimization of the reprojection function [90], where the sum of

TABLE IV
A LIST OF OPEN-SOURCE SOFTWARE PACKAGES FOR SPHERICAL CAMERA CALIBRATION.

Name	Language	Year	Website
OCamCalib	Matlab	2006	https://sites.google.com/site/scarabotix/ocamcalib-omnidirectional-camera-calibration-toolbox-for-matlab
	A toolbox to calibrate any central omnidirectional camera, i.e., panoramic cameras having a single effective viewpoint [79].		
Improved OcamCalib	Matlab	2015	https://github.com/urbste/ImprovedOcamCalib
	An add-on toolkit to the OCamCalib toolbox that implements calibration algorithms for wide-angle, fisheye, and omnidirectional cameras [81].		
LIBOMNICAL	Matlab	2014	https://www.cvlibs.net/projects/omnicam
	a MATLAB Toolbox to calibrate central and slightly non-central catadioptric cameras and catadioptric stereo setups [85].		
Omnidirectional Calibration Toolbox Mei	Matlab	2007	https://www.robots.ox.ac.uk/~cmei/Toolbox.html
	A toolbox implements the unified projection model to calibrate hyperbolic, parabolic, and folding mirrors and spherical and wide-angle sensors [77].		
camodocal	C++	2013	https://github.com/hengli/camodocal
	A toolbox for automatic intrinsic and extrinsic calibration of a camera rig with multiple generic cameras and odometry [78].		
kalibr	C++	2016	https://github.com/ethz-asl/kalibr
	A toolbox for multi-camera calibration and multi-sensor integration [86].		

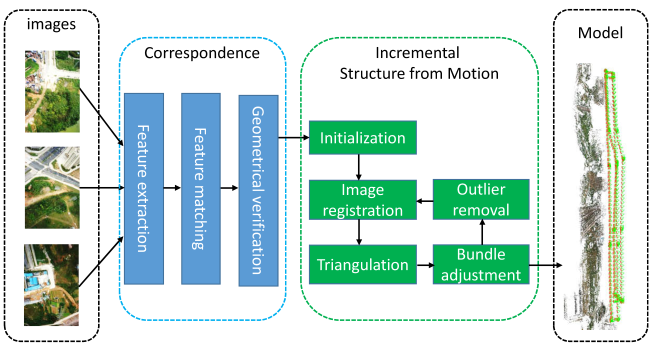

Fig. 14. The workflow of the incremental SfM [72].

errors between tie-point projections and their corresponding image points is minimized. The object function of BA is presented by the Equation (1).

$$\min_{C_j,X_i} \sum_{i=1}^{n} \sum_{j=1}^{m} \rho_{ij} \parallel P(C_j, X_i) - x_{ij} \parallel^2 \quad (1)$$

where X_i and C_j indicate a 3D point and a camera, respectively; $P(C_j, X_i)$ is the projection of point X_i on camera C_j; x_{ij} is an observed image point; $\|\bullet\|$ denotes L2-norm; ρ_{ij} is an indicator function with $\rho_{ij} = 1$ if point X_i is visible in camera C_j; otherwise $\rho_{ij} = 0$.

(2) Perspective SfM to spherical SfM

For spherical image orientation, there have been some attempts to adapt perspective SfM to spherical SfM in the last two decades. [91] proposed a two-step SfM for omnidirectional images depending on linear initialization and nonlinear optimization, which only recovers camera relative geometry. Uncertainty analysis and result comparison were also conducted and compared with perspective SfM. In the work of [92], both two-view and three-view geometry of spherical images have been analyzed and discussed. The epipolar geometry forms the basis for 3D reconstruction from spherical images, e.g., the pose recovery in a virtual navigation system [93].

With the usage of advanced MMS systems for urban city modeling and navigation, some researchers moved their attention to 3D reconstruction of large-scale scenes instead of two or three-view geometry estimation in the earlier work [94]. As one of the early pioneering attempts, [44] proposed an incremental SfM system that combines state-of-the-art techniques, e.g., local feature-based feature detection, approximate nearest neighbor-based feature matching, and robust essential matrix estimation and optimization. The performance of the proposed SfM system has been verified by using Google Street View images that cover a large street block [24], as presented in Fig. 15(a). For full spherical images, [17] investigated different error metrics on relative and absolute pose estimation and also designed the error approximations to reduce computational costs. These error metrics consist of the basic blocks for spherical SfM.

In contrast to professional sensors, recent years also witnessed the explosive development of consumer-grade spherical cameras [10] and their usage in 3D modeling. In [95], the von Mises-Fisher distribution was utilized to model the noise distribution of feature point positions on spherical images and to reformulate the error function in the bundle adjustment optimization. Meanwhile, spherical-n-point and triangulation algorithms that are suitable for spherical images have been proposed to achieve spherical video orientation and viewing direction stabilization. [26] embedded spherical cameras into a UAV platform and proposed a solution for panoramic photogrammetry, as presented in Fig. 15(b). In data processing, original spherical images are first reprojected to cubic images, which are then reconstructed by using existing commercial software. The study verifies the validation of spherical cameras for oblique photogrammetric 3D modeling of urban buildings.

Although extensive research has been reported in the liter-